Ingress Tool Transfer

Adversaries may transfer tools or other files from an external system into a compromised environment. Tools or files may be copied from an external adversary-controlled system to the victim network through the command and control channel or through alternate protocols such as ftp. Once present, adversaries may also transfer/spread tools between victim devices within a compromised environment (i.e. Lateral Tool Transfer).

On Windows, adversaries may use various utilities to download tools, such as `copy`, `finger`, certutil, and PowerShell commands such as `IEX(New-Object Net.WebClient).downloadString()` and `Invoke-WebRequest`. On Linux and macOS systems, a variety of utilities also exist, such as `curl`, `scp`, `sftp`, `tftp`, `rsync`, `finger`, and `wget`.[1] A number of these tools, such as `wget`, `curl`, and `scp`, also exist on ESXi. After downloading a file, a threat actor may attempt to verify its integrity by checking its hash value (e.g., via `certutil -hashfile`).[2]

Adversaries may also abuse installers and package managers, such as `yum` or `winget`, to download tools to victim hosts. Adversaries have also abused file application features, such as the Windows `search-ms` protocol handler, to deliver malicious files to victims through remote file searches invoked by User Execution (typically after interacting with Phishing lures).[3]

Files can also be transferred using various Web Services as well as native or otherwise present tools on the victim system.[4] In some cases, adversaries may be able to leverage services that sync between a web-based and an on-premises client, such as Dropbox or OneDrive, to transfer files onto victim systems. For example, by compromising a cloud account and logging into the service's web portal, an adversary may be able to trigger an automatic syncing process that transfers the file onto the victim's machine.[5]

ID: T1105

Sub-techniques: No sub-techniques

ⓘ Tactic: Command and Control

ⓘ Platforms: ESXi, Linux, Network Devices, Windows, macOS

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Procedure Examples

ID	Name	Description
C0028	2015 Ukraine Electric Power Attack	During the 2015 Ukraine Electric Power Attack, Sandworm Team pushed additional malicious tools onto an infected system to steal user credentials, move laterally, and destroy data. [6]
C0063	2025 Poland Wiper Attacks	During the 2025 Poland Wiper Attacks, the adversaries downloaded malicious payloads to the victim server. [7]
S0469	ABK	ABK has the ability to download files from C2.[8]
S1028	Action RAT	Action RAT has the ability to download additional payloads onto an infected machine.[9]
S0331	Agent Tesla	Agent Tesla can download additional files for execution on the victim's machine.[10][11]
S0092	Agent.btz	Agent.btz attempts to download an encrypted binary from a specified domain.[12]
G0130	Ajax Security Team	Ajax Security Team has used Wrapper/Gholee, custom-developed malware, which downloaded additional malware to the infected system.[13]
S1025	Amadey	Amadey can download and execute files to further infect a host machine with additional malware.[14]
S0504	Anchor	Anchor can download additional payloads.[15][16]
G0138	Andariel	Andariel has downloaded additional tools and malware onto compromised hosts.[17]
S1074	ANDROMEDA	ANDROMEDA can download additional payloads from C2.[18]
G0099	APT-C-36	APT-C-36 has downloaded binary data from a specified domain after the malicious document is opened.[19]
G0026	APT18	APT18 can upload a file to the victim's machine.[20]
G0007	APT28	APT28 has downloaded additional files, including by using a first-stage downloader to contact the C2 server to obtain the second-stage implant.[21][22][23][24][25]
G0016	APT29	APT29 has downloaded additional tools and malware onto compromised networks.[26][27][28][29]
G0022	APT3	APT3 has a tool that can copy files to remote machines.[30]
G0050	APT32	APT32 has added JavaScript to victim websites to download additional frameworks that profile and compromise website visitors.[31]
G0064	APT33	APT33 has downloaded additional files and programs from its C2 server.[32][33]
G0067	APT37	APT37 has downloaded second stage malware from compromised websites.[34][35][36][37]
G0082	APT38	APT38 used a backdoor, NESTEGG, that has the capability to download and upload files to and from a victim's machine.[38] Additionally, APT38 has downloaded other payloads onto a victim's machine.[39]
G0087	APT39	APT39 has downloaded tools to compromised hosts.[40][41]
G0096	APT41	APT41 used certutil to download additional files.[42][43][44] APT41 downloaded post-exploitation tools such as Cobalt Strike via command shell following initial access.[45] APT41 has uploaded Procdump and NATBypass to a staging directory and has used these tools in follow-on activities.[46]

ID	Name	Description
S0373	Astaroth	Astaroth uses certutil and BITSAdmin to download additional malware. [51][52][53]
S1087	AsyncRAT	AsyncRAT has the ability to download files including over SFTP.[54][55]
S0438	Attor	Attor can download additional plugins, updates and other files. [56]
S0347	AuditCred	AuditCred can download files and additional malware.[57]
S0473	Avenger	Avenger has the ability to download files from C2 to a compromised host.[8]
S0344	Azorult	Azorult can download and execute additional files. Azorult has also downloaded a ransomware payload called Hermes.[58][59]
S0414	BabyShark	BabyShark has downloaded additional files from the C2.[60][61]
S0475	BackConfig	BackConfig can download and execute additional payloads on a compromised host.[62]
S0093	Backdoor.Oldrea	Backdoor.Oldrea can download additional modules from C2.[63]
G0135	BackdoorDiplomacy	BackdoorDiplomacy has downloaded additional files and tools onto a compromised host.[64]
S0642	BADFLICK	BADFLICK has download files from its C2 server.[65]
S1081	BADHATCH	BADHATCH has the ability to load a second stage malicious DLL file onto a compromised machine.[66]
S0128	BADNEWS	BADNEWS is capable of downloading additional files through C2 channels, including a new version of itself. [67][68][69]
S0337	BadPatch	BadPatch can download and execute or update malware.[70]
S0234	Bandook	Bandook can download files to the system.[71]
S0239	Bankshot	Bankshot uploads files and secondary payloads to the victim's machine.[72]
S0534	Bazar	Bazar can download and deploy additional payloads, including ransomware and post-exploitation frameworks such as Cobalt Strike.[73][74][75][76]
S0470	BBK	BBK has the ability to download files from C2 to the infected host.[8]
S1246	BeaverTail	BeaverTail has been used to download a malicious payload to include Python based malware InvisibleFerret. [77][78][79][80][81][82]
S0574	BendyBear	BendyBear is designed to download an implant from a C2 server.[83]
S0017	BISCUIT	BISCUIT has a command to download a file from the C2 server.[84]
S0268	Bisonal	Bisonal has the capability to download files to execute on the victim's machine.[85][86][87]
S0190	BITSAdmin	BITSAdmin can be used to create BITS Jobs to upload and/or download files.[88]
G1002	BITTER	BITTER has downloaded additional malware and tools onto a compromised host.[89][90]
G1043	BlackByte	BlackByte has transferred tools such as Cobalt Strike to victim environments from file sharing and hosting websites.[91]

ID	Name	Description
S0360	BONDUPDATER	BONDUPDATER can download or upload files from its C2 server.[95]
S0635	BoomBox	BoomBox has the ability to download next stage malware components to a compromised system.[96]
S0651	BoxCaon	BoxCaon can download files.[97]
S0204	Briba	Briba downloads files onto infected hosts.[98]
S9015	BRICKSTORM	BRICKSTORM has the ability to download files from the Adversaries C2 server to the compromised system.[99][100][101][102]
G0060	BRONZE BUTLER	BRONZE BUTLER has used various tools to download files, including DGet (a similar tool to wget).[103]
S1063	Brute Ratel C4	Brute Ratel C4 can download files to compromised hosts.[104][105]
S0471	build_downer	build_downer has the ability to download files from C2 to the infected host.[8]
S1039	Bumblebee	Bumblebee can download and execute additional payloads including through the use of a `Dex` command.[106][107][108]
S0482	Bundlore	Bundlore can download and execute new versions of itself.[109]
S1118	BUSHWALK	BUSHWALK can write malicious payloads sent through a web request's command parameter.[110][111]
C0010	C0010	During C0010, UNC3890 actors downloaded tools and malware onto a compromised host.[112]
C0015	C0015	During C0015, the threat actors downloaded additional tools and files onto a compromised network.[113]
C0017	C0017	During C0017, APT41 downloaded malicious payloads onto compromised systems.[114]
C0018	C0018	During C0018, the threat actors downloaded additional tools, such as Mimikatz and Sliver, as well as Cobalt Strike and AvosLocker ransomware onto the victim network.[115][116]
C0021	C0021	During C0021, the threat actors downloaded additional tools and files onto victim machines.[117][118]
C0026	C0026	During C0026, the threat actors downloaded malicious payloads onto select compromised hosts.[18]
C0027	C0027	During C0027, Scattered Spider downloaded tools using victim organization systems.[119]
S0274	Calisto	Calisto has the capability to upload and download files to the victim's machine.[120]
S0077	CallMe	CallMe has the capability to download a file to the victim from the C2 server.[121]
S9016	Caminho	Caminho has the ability to download files onto compromised hosts.[122]
S9042	CanisterWorm	CanisterWorm has downloaded and executed binaries from dead drop URLs retrieved from ICP canister C2 nodes.[123][124] CanisterWorm has also downloaded kubectl to compromised environments if it was not already installed.[123]
S0351	Cannon	Cannon can download a payload for execution.[125]
S0484	Carberp	Carberp can download and execute new plugins from the C2 server. [126][127]
S0348	Cardinal RAT	Cardinal RAT can download and execute additional payloads.[128]

ID	Name	Description
S0160	certutil	certutil can be used to download files from a given URL.[133][134]
S0631	Chaes	Chaes can download additional files onto an infected machine.[135]
S0674	CharmPower	CharmPower has the ability to download additional modules to a compromised host.[136]
S0144	ChChes	ChChes is capable of downloading files, including additional modules.[137][138][139]
G0114	Chimera	Chimera has remotely copied tools and malware onto targeted systems.[140]
S1149	CHIMNEYSWEEP	CHIMNEYSWEEP can download additional files from C2.[141]
S0020	China Chopper	China Chopper's server component can download remote files.[142][143][144][145][146]
S0023	CHOPSTICK	CHOPSTICK is capable of performing remote file transmission.[147]
S0667	Chrommme	Chrommme can download its code from C2.[148]
G1021	Cinnamon Tempest	Cinnamon Tempest has downloaded files, including Cobalt Strike, to compromised hosts.[149]
S0054	CloudDuke	CloudDuke downloads and executes additional malware from either a Web address or a Microsoft OneDrive account.[28]
S0106	cmd	cmd can be used to copy files to/from a remotely connected external system.[150]
G0080	Cobalt Group	Cobalt Group has used public sites such as github.com and sendspace.com to upload files and then download them to victim computers.[151][4] The group's JavaScript backdoor is also capable of downloading files.[152]
S0154	Cobalt Strike	Cobalt Strike can deliver additional payloads to victim machines.[153][154]
S0369	CoinTicker	CoinTicker executes a Python script to download its second stage.[155]
S0608	Conficker	Conficker downloads an HTTP server to the infected machine.[156]
G0142	Confucius	Confucius has downloaded additional files and payloads onto a compromised host following initial access.[157][158]
S0492	CookieMiner	CookieMiner can download additional scripts from a web server.[159]
S0137	CORESHELL	CORESHELL downloads another dropper from its C2 server.[160]
S0614	CostaBricks	CostaBricks has been used to load SombRAT onto a compromised host.[161]
C0004	CostaRicto	During CostaRicto, the threat actors downloaded malware and tools onto a compromised host.[161]
S1023	CreepyDrive	CreepyDrive can download files to the compromised host.[162]
S0115	Crimson	Crimson contains a command to retrieve files from its C2 server.[163][164][165]
S0498	Cryptoistic	Cryptoistic has the ability to send and receive files.[166]
S0527	CSPY Downloader	CSPY Downloader can download additional tools to a compromised host.[167]

ID	Name	Description
G1034	Daggerfly	Daggerfly has used PowerShell and BITSAdmin to retrieve follow-on payloads from external locations for execution on victim machines.[173]
S1014	DanBot	DanBot can download additional files to a targeted system.[174]
S0334	DarkComet	DarkComet can load any files onto the infected machine to execute.[175][176]
S1111	DarkGate	DarkGate retrieves cryptocurrency mining payloads and commands in encrypted traffic from its command and control server.[177] DarkGate uses Windows Batch scripts executing the `curl` command to retrieve follow-on payloads.[178] DarkGate has stolen `sitemanager.xml` and `recentservers.xml` from `%APPDATA%\FileZilla\` if present.[179]
G0012	Darkhotel	Darkhotel has used first-stage payloads that download additional malware from C2 servers.[180]
S1066	DarkTortilla	DarkTortilla can download additional packages for keylogging, cryptocurrency mining, and other capabilities; it can also retrieve malicious payloads such as Agent Tesla, AsyncRat, NanoCore, RedLine, Cobalt Strike, and Metasploit.[181]
S0187	Daserf	Daserf can download remote files.[182][103]
S0255	DDKONG	DDKONG downloads and uploads files on the victim's machine.[183]
S0616	DEATHRANSOM	DEATHRANSOM can download files to a compromised host.[184]
S0354	Denis	Denis deploys additional backdoors and hacking tools to the system.[185]
S0659	Diavol	Diavol can receive configuration updates and additional payloads including wscpy.exe from C2.[186]
S0200	Dipsind	Dipsind can download remote files.[187]
S1088	Disco	Disco can download files to targeted systems via SMB.[188]
S1021	DnsSystem	DnsSystem can download files to compromised systems after receiving a command with the string `downloaddd`.[189]
S0213	DOGCALL	DOGCALL can download and execute additional payloads.[190]
S0600	Doki	Doki has downloaded scripts from C2.[191]
S0695	Donut	Donut can download and execute previously staged shellcode payloads.[192]
S0472	down_new	down_new has the ability to download files to the compromised host.[8]
S0134	Downdelph	After downloading its main config file, Downdelph downloads multiple payloads from C2 servers.[193]
S9021	DOWNIISSA	DOWNIISSA can download files to the compromised host.[194]
G0035	Dragonfly	Dragonfly has copied and installed tools for operations once in the victim environment.[195]
S0694	DRATzarus	DRATzarus can deploy additional tools onto an infected machine.[196]
S0547	DropBook	DropBook can download and execute additional files.[197][198]
S0502	Drovorub	Drovorub can download files to a compromised host.[199]

ID	Name	Description
S0554	Egregor	Egregor has the ability to download files from its C2 server.[204][205]
G0066	Elderwood	The Ritsol backdoor trojan used by Elderwood can download files onto a compromised host from a remote location.[206]
S0081	Elise	Elise can download additional files from the C2 server for execution.[207]
S0082	Emissary	Emissary has the capability to download files from the C2 server.[208]
S0367	Emotet	Emotet can download follow-on payloads and items via malicious `url` parameters in obfuscated PowerShell code.[209]
S0363	Empire	Empire can upload and download to and from a victim machine.[210]
S0404	esentutl	esentutl can be used to copy files from a given URL.[211]
S0396	EvilBunny	EvilBunny has downloaded additional Lua scripts from the C2.[212]
S0568	EVILNUM	EVILNUM can download and upload files to the victim's computer.[213][214]
G0120	Evilnum	Evilnum can deploy additional components or tools as needed.[213]
S0401	Exaramel for Linux	Exaramel for Linux has a command to download a file from and to a remote C2 server.[215][216]
S0569	Explosive	Explosive has a function to download a file to the infected system.[217]
S0171	Felismus	Felismus can download files from remote servers.[218]
S0267	FELIXROOT	FELIXROOT downloads and uploads files to and from the victim's machine.[219][220]
G1016	FIN13	FIN13 has downloaded additional tools and malware to compromised systems.[221][222]
G0046	FIN7	FIN7 has downloaded additional malware to execute on the victim's machine, including by using a PowerShell script to launch shellcode that retrieves an additional payload.[223][224][225][226]
G0061	FIN8	FIN8 has used remote code execution to download subsequent payloads.[227][228]
S0696	Flagpro	Flagpro can download additional malware from the C2 server.[229]
S0381	FlawedAmmyy	FlawedAmmyy can transfer files from C2.[230]
S0661	FoggyWeb	FoggyWeb can receive additional malicious components from an actor controlled C2 server and execute them on a compromised AD FS server.[231]
G0117	Fox Kitten	Fox Kitten has downloaded additional tools including PsExec directly to endpoints.[232]
C0001	Frankenstein	During Frankenstein, the threat actors downloaded files and tools onto a victim machine.[233]
S0095	ftp	ftp may be abused by adversaries to transfer tools or files from an external system into a compromised environment.[234][235]
S1044	FunnyDream	FunnyDream can download additional files onto a compromised host.[236]
C0007	FunnyDream	During FunnyDream, the threat actors downloaded additional droppers and backdoors onto a compromised

ID	Name	Description
G0047	Gamaredon Group	Gamaredon Group has downloaded additional malware and tools onto a compromised host.[238][239][240][241][242][243] For example, Gamaredon Group uses a backdoor script to retrieve and decode additional payloads once in victim environments.[244]
S0168	Gazer	Gazer can execute a task to download a file.[245][246]
S0666	Gelsemium	Gelsemium can download additional plug-ins to a compromised host.[148]
S0032	gh0st RAT	gh0st RAT can download files to the victim's machine.[247][248]
S9010	GlassWorm	GlassWorm has downloaded additional payloads from C2.[249][250][251][252]
S0249	Gold Dragon	Gold Dragon can download additional components from the C2 server.[253]
S0493	GoldenSpy	GoldenSpy constantly attempts to download and execute files from the remote C2, including GoldenSpy itself if not found on the system.[254]
S0588	GoldMax	GoldMax can download and execute additional files.[255][256]
S1138	Gootloader	Gootloader can fetch second stage code from hardcoded web domains.[257][258]
G0078	Gorgon Group	Gorgon Group malware can download additional files from C2 servers.[259]
S0531	Grandoreiro	Grandoreiro can download its second stage from a hardcoded URL within the loader's code.[260][261]
S0342	GreyEnergy	GreyEnergy can download additional modules and payloads.[220]
S0632	GrimAgent	GrimAgent has the ability to download and execute additional payloads.[262]
S0561	GuLoader	GuLoader can download further malware for execution on the victim's machine.[263]
S0132	H1N1	H1N1 contains a command to download and execute a file from a remotely hosted URL using WinINet HTTP requests.[264]
G0125	HAFNIUM	HAFNIUM has downloaded malware and tools--including Nishang and PowerCat--onto a compromised host.[265][145]
S0499	Hancitor	Hancitor has the ability to download additional files from C2.[266]
S1211	Hannotog	Hannotog can download additional files to the victim machine.[267]
S0214	HAPPYWORK	can download and execute a second-stage payload.[34]
S1229	Havoc	Havoc has the ability to upload files to infected systems.[268][269]
S0170	Helminth	Helminth can download additional files.[270]
G1001	HEXANE	HEXANE has downloaded additional payloads and malicious scripts onto a compromised host.[271]
S1249	HexEval Loader	HexEval Loader has been used to download a malicious payload to include BeaverTail.[272][78][79]
S0087	Hi-Zor	Hi-Zor has the ability to upload and download files from its C2 server.[273]
S9023	HiddenFace	HiddenFace can download files from the C2 to victim systems.[274][275]

ID	Name	Description
C0038	HomeLand Justice	During HomeLand Justice, threat actors used web shells to download files to compromised infrastructure.[279]
S0376	HOPLIGHT	HOPLIGHT has the ability to connect to a remote host in order to upload and download files.[280]
S0431	HotCroissant	HotCroissant has the ability to upload a file from the command and control (C2) server to the victim machine.[281]
S0070	HTTPBrowser	HTTPBrowser is capable of writing a file to the compromised system from the C2 server.[282]
S9007	HTTPTroy	HTTPTroy has the ability to download files from C2 using the `down <FILENAME>` command.[283]
S0203	Hydraq	Hydraq creates a backdoor through which remote attackers can download files and additional malware components.[284][285]
S0398	HyperBro	HyperBro has the ability to download additional files.[286]
S0483	IcedID	IcedID has the ability to download additional modules and a configuration file from C2.[287][288][289][290]
S1152	IMAPLoader	IMAPLoader is a loader used to retrieve follow-on payload encoded in email messages for execution on victim systems.[291]
G1032	INC Ransom	INC Ransom has downloaded tools to compromised servers including Advanced IP Scanner. [292][293]
G0136	IndigoZebra	IndigoZebra has downloaded additional files and tools from its C2 server.[97]
G0119	Indrik Spider	Indrik Spider has downloaded additional scripts, malware, and tools onto a compromised host.[294][295][296]
S0604	Industroyer	Industroyer downloads a shellcode payload from a remote C2 server and loads it into memory.[297]
S1245	InvisibleFerret	InvisibleFerret has downloaded "AnyDesk.exe" into the user's home directory from the C2 server when checks for the service fail to identify its presence in the victim environment.[80] InvisibleFerret has also been configured to download additional payloads using a command which calls to the /bow URI.[298][81]
S0260	InvisiMole	InvisiMole can upload files to the victim's machine for operations.[299][300]
S0015	Ixeshe	Ixeshe can download and execute additional files.[301]
S0528	Javali	Javali can download payloads from remote C2 servers.[53]
S0044	JHUHUGIT	JHUHUGIT can retrieve an additional payload from its C2 server.[302][303] JHUHUGIT has a command to download files to the victim's machine.[304]
S0201	JPIN	JPIN can download files and upgrade itself.[187]
S0283	jRAT	jRAT can download and execute files.[305][306][307]
S0648	JSS Loader	JSS Loader has the ability to download malicious executables to a compromised host.[308]
S0215	KARAE	KARAE can upload and download files, including second-stage malware.[34]
S0088	Kasidet	Kasidet has the ability to download and execute additional files.[309]
S0265	Kazuar	Kazuar downloads additional plug-ins to load on the victim's machine, including the ability to upgrade and

ID	Name	Description
S1020	Kevin	Kevin can download files to the compromised host.[271]
S0387	KeyBoy	KeyBoy has a download and upload functionality.[313][314]
S0271	KEYMARBLE	KEYMARBLE can upload files to the victim's machine and can download additional payloads.[315]
S0526	KGH_SPY	KGH_SPY has the ability to download and execute code from remote servers.[167]
G0094	Kimsuky	Kimsuky has downloaded additional scripts, tools, and malware onto victim systems.[316][43][317][318]
S0599	Kinsing	Kinsing has downloaded additional lateral movement scripts from C2.[319]
S0437	Kivars	Kivars has the ability to download and execute files.[320]
S0250	Koadic	Koadic can download additional files and tools.[321][322]
S0669	KOCTOPUS	KOCTOPUS has executed a PowerShell command to download a file to the system.[322]
S0356	KONNI	KONNI can download files and execute them on the victim's machine.[323][324]
C0035	KV Botnet Activity	KV Botnet Activity included the use of scripts to download additional payloads when compromising network nodes.[325]
S0236	Kwampirs	Kwampirs downloads additional files from C2 servers.[326]
S1160	Latrodectus	Latrodectus can download and execute PEs, DLLs, and shellcode from C2.[290][327][328]
G0032	Lazarus Group	Lazarus Group has downloaded files, malware, and tools from its C2 onto a compromised host.[329][330][331][166][172][332][333][334][335][336]
G0140	LazyScripter	LazyScripter had downloaded additional tools to a compromised host.[322]
G0065	Leviathan	Leviathan has downloaded additional scripts and files from adversary-controlled servers.[337][142]
S0395	LightNeuron	LightNeuron has the ability to download and execute additional files.[338]
S1185	LightSpy	On macOS, LightSpy downloads a `.json` file from the C2 server. The `.json` file contains metadata about the plugins to be downloaded, including their URL, name, version, and MD5 hash. LightSpy retrieves the plugins specified in the `.json` file, which are compiled `.dylib` files. These `.dylib` files provide task and platform specific functionality. LightSpy also imports open-source libraries to manage socket connections.[339]
S0211	Linfo	Linfo creates a backdoor through which remote attackers can download files onto compromised hosts.[340]
S0513	LiteDuke	LiteDuke has the ability to download files.[341]
S0680	LitePower	LitePower has the ability to download payloads containing system commands to a compromised host.[342]
S0681	Lizar	Lizar can download additional plugins, files, and tools.[343][344][345]
S9020	LODEINFO	LODEINFO has the ability to download additional files from the C2.[346][347][348]
S0447	Lokibot	Lokibot downloaded several staged items onto the victim's machine.[349]

ID	Name	Description
G1014	LuminousMoth	LuminousMoth has downloaded additional malware and tools onto a compromised host.[353][354]
S0409	Machete	Machete can download additional files for execution on the victim's machine.[355]
S1016	MacMa	MacMa has downloaded additional files, including an exploit for used privilege escalation.[356][357]
S1048	macOS.OSAMiner	macOS.OSAMiner has used `curl` to download a Stripped Payloads from a public facing adversary-controlled webpage.
S1060	Mafalda	Mafalda can download additional files onto the compromised host.[358]
G0059	Magic Hound	Magic Hound has downloaded additional code and files from servers onto victims.[359][360][361][362]
S1182	MagicRAT	MagicRAT can import and execute additional payloads.[363]
S0652	MarkiRAT	MarkiRAT can download additional files and tools from its C2 server, including through the use of BITSAdmin.[364]
S0500	MCMD	MCMD can upload additional files to a compromised host.[365]
S0459	MechaFlounder	MechaFlounder has the ability to upload and download files to and from a compromised host.[366]
G1051	Medusa Group	Medusa Group has leveraged certutil, PowerShell, and Windows Command to download additional tools to include RMM services.[367] Medusa Group has also engaged in "Bring Your Own Vulnerable Driver" (BYOVD) and downloaded vulnerable or signed drivers to the victim environment to disable security tools.[367][368]
S0530	Melcoz	Melcoz has the ability to download additional files to a compromised host.[53]
G0045	menuPass	menuPass has installed updates and new malware on victims.[369][370]
G1013	Metador	Metador has downloaded tools and malware onto a compromised system.[371]
S1059	metaMain	metaMain can download files onto compromised systems.[371][358]
S0455	Metamorfo	Metamorfo has used MSI files to download additional files to execute.[372][373][374][375]
S0688	Meteor	Meteor has the ability to download additional files for execution on the victim's machine.[376]
S0339	Micropsia	Micropsia can download and execute an executable from the C2 server.[377][378]
S1015	Milan	Milan has received files from C2 and stored them in log folders beginning with the character sequence `a9850d2f`.[379]
S9043	Mini Shai-Hulud	Mini Shai-Hulud has the ability to download additional payloads from adversary controlled or compromised infrastructure.[380][381][382]
S0051	MiniDuke	MiniDuke can download additional encrypted backdoors onto the victim via GIF files.[383][341]
S0084	Mis-Type	Mis-Type has downloaded additional malware and files onto a compromised host.[384]
S0083	Misdat	Misdat is capable of downloading files from the C2.[384]
S0080	Mivast	Mivast has the capability to download and execute .exe files.[385]

ID	Name	Description
G1036	Moonstone Sleet	Moonstone Sleet retrieved a final stage payload from command and control infrastructure during initial installation on victim systems.[389]
S0284	More_eggs	More_eggs can download and launch additional payloads.[390][391]
G1009	Moses Staff	Moses Staff has downloaded and installed web shells to following path `C:\inetpub\wwwroot\aspnet_client\system_web\IISpool.aspx`.[392]
S0256	Mosquito	Mosquito can upload and download files to the victim.[393]
S9032	MuddyViper	MuddyViper has the ability to download files from the C2 server. Additionally, MuddyViper has the ability to download a file in chunks with sleep time between each chunk.[394]
G0069	MuddyWater	MuddyWater has used malware that can upload additional files to the victim's machine.[395][396][397][398] MuddyWater has used PowerShell commands to install remote management and monitoring (RMM) software on the victim's machine to conduct espionage and to exfiltrate data.[399]
G0129	Mustang Panda	Mustang Panda has downloaded additional executables following the initial infection stage.[400][401][402][403] Mustang Panda has also leveraged Visual Studio Code `code.exe` and Dev Tunnels using `DevTunnel.exe` to propagate additional tools and payloads.[404]
G1020	Mustard Tempest	Mustard Tempest has deployed secondary payloads and third stage implants to compromised hosts.[405]
S0228	NanHaiShu	NanHaiShu can download additional files from URLs.[337]
S0336	NanoCore	NanoCore has the capability to download and activate additional modules for execution.[406][407]
S0247	NavRAT	NavRAT can download files remotely.[408]
S0272	NDiskMonitor	NDiskMonitor can download and execute a file from given URL.[69]
S0630	Nebulae	Nebulae can download files from C2.[409]
S1189	Neo-reGeorg	Neo-reGeorg has the ability to download files to targeted systems.[410]
S0691	Neoichor	Neoichor can download additional files onto a compromised host.[311]
S0210	Nerex	Nerex creates a backdoor through which remote attackers can download files onto a compromised host.[206]
S0457	Netwalker	Operators deploying Netwalker have used psexec and certutil to retrieve the Netwalker payload.[411]
S0198	NETWIRE	NETWIRE can downloaded payloads from C2 to the compromised host.[412][413]
S1192	NICECURL	NICECURL has the ability to download additional content onto an infected machine, e.g. by using `curl`.[414]
S0118	Nidiran	Nidiran can download and execute files.[415]
C0002	Night Dragon	During Night Dragon, threat actors used administrative utilities to deliver Trojan components to remote systems.[416]
S1090	NightClub	NightClub can load multiple additional plugins on an infected host.[188]
S0385	njRAT	njRAT can download files to the victim's machine.[417][418] APT-C-36 has used modified versions of njRAT to enable the download of .NET assemblies.[419]

ID	Name	Description
S1172	OilBooster	OilBooster can download and execute files from an actor-controlled OneDrive account.[424]
S1171	OilCheck	OilCheck can download staged payloads from an actor-controlled infrastructure.[424]
G0049	OilRig	OilRig had downloaded remote files onto victim infrastructure.[425][426]
S0439	Okrum	Okrum has built-in commands for uploading, downloading, and executing files to the system.[427]
S0264	OopsIE	OopsIE can download files from its C2 server to the victim's machine.[428][429]
C0022	Operation Dream Job	During Operation Dream Job, Lazarus Group downloaded multistage malware and tools onto a compromised host.[196][430][431]
C0006	Operation Honeybee	During Operation Honeybee, the threat actors downloaded additional malware and malicious scripts onto a compromised host.[432]
C0048	Operation MidnightEclipse	During Operation MidnightEclipse, threat actors downloaded additional payloads on compromised devices.[433][434]
C0013	Operation Sharpshooter	During Operation Sharpshooter, additional payloads were downloaded after a target was infected with a first-stage downloader.[435]
C0014	Operation Wocao	During Operation Wocao, threat actors downloaded additional files to the infected system.[436]
S0229	Orz	Orz can download files onto the victim.[337]
S0402	OSX/Shlayer	OSX/Shlayer can download payloads, and extract bytes from files. OSX/Shlayer uses the `curl -fsL "$url" >$tmp_path` command to download malicious payloads into a temporary directory.[437][438][439][440]
S0352	OSX_OCEANLOTUS.D	OSX_OCEANLOTUS.D has a command to download and execute a file on the victim's machine.[441][442]
C0042	Outer Space	During Outer Space, OilRig downloaded additional tools to comrpomised infrastructure.[443]
S1017	OutSteel	OutSteel can download files from its C2 server.[444]
S0598	P.A.S. Webshell	P.A.S. Webshell can upload and download files to and from compromised hosts.[216]
S0626	P8RAT	P8RAT can download additional payloads to a target system.[203]
S0664	Pandora	Pandora can load additional drivers and files onto a victim machine.[445]
S0208	Pasam	Pasam creates a backdoor through which remote attackers can upload files.[446]
G0040	Patchwork	Patchwork payloads download additional files from the C2 server.[447][69]
S0587	Penquin	Penquin can execute the command code `do_download` to retrieve remote files from C2.[448]
S0643	Peppy	Peppy can download and execute remote files.[163]
S9014	PHASEJAM	PHASEJAM has the ability to upload files onto the compromised appliance.[449]
S9028	PHPsert	PHPsert has the ability to retrieve remote payloads.[450]
S0501	PipeMon	PipeMon can install additional modules via C2 commands.[451]

ID	Name	Description
G1040	Play	Play has used Cobalt Strike to download files to compromised machines.[454]
S0435	PLEAD	PLEAD has the ability to upload and download files to and from an infected host.[455]
S0013	PlugX	PlugX has a module to download and execute files on the compromised machine.[456][457][458][459]
S0428	PoetRAT	PoetRAT has the ability to copy files and download/upload files into C2 channels using FTP and HTTPS.[460][461]
S0012	PoisonIvy	PoisonIvy creates a backdoor through which remote attackers can upload files.[462]
S0518	PolyglotDuke	PolyglotDuke can retrieve payloads from the C2 server.[341]
S0453	Pony	Pony can download additional files onto the infected system.[463]
S0150	POSHSPY	POSHSPY downloads and executes additional PowerShell code and Windows binaries.[464]
S0139	PowerDuke	PowerDuke has a command to download a file.[465]
S1173	PowerExchange	PowerExchange can decode Base64-encoded files and call `WriteAllBytes` to write the files to compromised hosts.[466]
S1012	PowerLess	PowerLess can download additional payloads to a compromised host.[467]
S0685	PowerPunch	PowerPunch can download payloads from adversary infrastructure.[241]
S0145	POWERSOURCE	POWERSOURCE has been observed being used to download TEXTMATE and the Cobalt Strike Beacon payload onto victims.[468]
S0223	POWERSTATS	POWERSTATS can retrieve and execute additional PowerShell payloads from the C2 server.[469]
S0184	POWRUNER	POWRUNER can download or upload files from its C2 server.[425]
S0613	PS1	CostaBricks can download additional payloads onto a compromised host.[161]
S0078	Psylo	Psylo has a command to download a file to the system from its C2 server.[121]
S0147	Pteranodon	Pteranodon can download and execute additional files.[238][470][471]
S1228	PUBLOAD	PUBLOAD has acted as a stager that can download the next-stage payload from its C2 server.[472][473][474][475][476] PUBLOAD has also delivered FDMTP as a secondary control tool and PTSOCKET for exfiltration to some infected systems.[477]
S0196	PUNCHBUGGY	PUNCHBUGGY can download additional files and payloads to compromised hosts.[478][479]
S0192	Pupy	Pupy can upload and download to/from a victim machine.[480]
S9019	PureCrypter	PureCrypter can download additional payloads for execution on the compromised host.[481][482]
S0650	QakBot	QakBot has the ability to download additional components and malware.[483][484][485][486][487][488]
C0055	Quad7 Activity	Quad7 Activity has downloaded additional binaries from a remote File Transfer Protocol (FTP) server to compromised devices.[489]

ID	Name	Description
G0075	Rancor	Rancor has downloaded additional malware, including by using certutil.[183]
S0055	RARSTONE	RARSTONE downloads its backdoor component from a C2 server and loads it directly into memory.[494]
S1130	Raspberry Robin	Raspberry Robin retrieves its second stage payload in a variety of ways such as through msiexec.exe abuse, or running the curl command to download the payload to the victim's `%AppData%` folder.[495][496]
S0241	RATANKBA	RATANKBA uploads and downloads information.[497][498]
S0662	RCSession	RCSession has the ability to drop additional files to an infected machine.[499]
S0495	RDAT	RDAT can download files via DNS.[500]
S0153	RedLeaves	RedLeaves is capable of downloading a file from a specified URL.[501]
S1240	RedLine Stealer	RedLine Stealer has the ability download additional payloads.[502][503]
C0056	RedPenguin	During RedPenguin, UNC3886 used backdoor malware capable of downloading files to compromised infrastructure.[504]
S0511	RegDuke	RegDuke can download files from C2.[341]
S1187	reGeorg	reGeorg has the ability to download files to targeted systems.[410]
S0332	Remcos	Remcos can upload and download files to and from the victim's machine.[505][506]
S0166	RemoteCMD	RemoteCMD copies a file over to the remote system before execution.[507]
S0592	RemoteUtilities	RemoteUtilities can upload and download files to and from a target machine.[398]
S0125	Remsec	Remsec contains a network loader to receive executable modules from remote attackers and run them on the local victim. It can also upload and download files over HTTP and HTTPS.[508][509]
S0379	Revenge RAT	Revenge RAT has the ability to upload and download files.[510]
S0496	REvil	REvil can download a copy of itself from an attacker controlled IP address to the victim machine.[511][512][513]
S0258	RGDoor	RGDoor uploads and downloads files to and from the victim's machine.[514]
S1222	RIFLESPINE	RIFLESPINE can download and execute files.[515]
G0106	Rocke	Rocke used malware to download additional malicious files to the target system.[516]
S0270	RogueRobin	RogueRobin can save a new file to the system from the C2 server.[517][518]
S0240	ROKRAT	ROKRAT can retrieve additional malicious payloads from its C2 server.[519][520][37][521]
S0148	RTM	RTM can download additional files.[522][523]
S0085	S-Type	S-Type can download additional files onto a compromised host.[384]
S1018	Saint Bot	Saint Bot can download additional files onto a compromised host.[444]
S0074	Sakula	Sakula has the capability to download files.[524]

ID	Name	Description
S1085	Sardonic	Sardonic has the ability to upload additional malicious files to a compromised machine.[527]
G1015	Scattered Spider	Scattered Spider has downloaded the Teleport remote access tool to compromised VMware vCenter Servers.[528]
S0461	SDBbot	SDBbot has the ability to download a DLL from C2 to a compromised host.[529]
S0053	SeaDuke	SeaDuke is capable of uploading and downloading files.[530]
S0345	Seasalt	Seasalt has a command to download additional files.[84][84]
S0185	SEASHARPEE	SEASHARPEE can download remote files onto victims.[531]
S0382	ServHelper	ServHelper may download additional files to execute.[532][533]
S0639	Seth-Locker	Seth-Locker has the ability to download and execute files on a compromised host.[534]
S0596	ShadowPad	ShadowPad has downloaded code from a C2 server.[535]
C0045	ShadowRay	During ShadowRay, threat actors downloaded and executed the XMRig miner on targeted hosts.[536]
S9008	Shai-Hulud	Shai-Hulud has downloaded packages from code repositories.[537][538][539][540] Shai-Hulud has also downloaded and executed the secrets-discovery tool TruffleHog to gather sensitive data.[541][538][542][539][540]
S0140	Shamoon	Shamoon can download an executable to run on the victim.[543]
C0058	SharePoint ToolShell Exploitation	During SharePoint ToolShell Exploitation, threat actors used a loader to download and execute ransomware.[544]
S1019	Shark	Shark can download additional files from its C2 via HTTP or DNS.[379][545]
S1089	SharpDisco	SharpDisco has been used to download a Python interpreter to `C:\Users\Public\WinTN\WinTN.exe` as well as other plugins from external sources.[188]
S0546	SharpStage	SharpStage has the ability to download and execute additional payloads via a DropBox API.[197][198]
S0450	SHARPSTATS	SHARPSTATS has the ability to upload and download files.[546]
S0444	ShimRat	ShimRat can download additional files.[547]
S0445	ShimRatReporter	ShimRatReporter had the ability to download additional payloads.[547]
G1057	ShinyHunters	ShinyHunters has deployed custom scripts to targeted systems from customized MeshAgents in their staging environment.[548]
S0217	SHUTTERSPEED	SHUTTERSPEED can download and execute an arbitary executable.[34]
S0589	Sibot	Sibot can download and execute a payload onto a compromised system.[255]
G1008	SideCopy	SideCopy has delivered trojanized executables via spearphishing emails that contacts actor-controlled servers to download malicious payloads.[9]
S0610	SideTwist	SideTwist has the ability to download additional files.[549]

ID	Name	Description
S1110	SLIGHTPULSE	RAPIDPULSE can transfer files to and from compromised hosts.[555]
S0633	Sliver	Sliver can download additional content and files from the Sliver server to the client residing on the victim machine using the `upload` command.[556][557]
S0533	SLOTHFULMEDIA	SLOTHFULMEDIA has downloaded files onto a victim machine.[558]
S0218	SLOWDRIFT	SLOWDRIFT downloads additional payloads.[34]
S1035	Small Sieve	Small Sieve has the ability to download files.[559]
S0226	Smoke Loader	Smoke Loader downloads a new version of itself once it has installed. It also downloads additional plugins.[560]
S0649	SMOKEDHAM	SMOKEDHAM has used Powershell to download UltraVNC and ngrok from third-party file sharing sites.[561]
S1086	Snip3	Snip3 can download additional payloads to compromised systems.[562][563]
S1124	SocGholish	SocGholish can download additional malware to infected hosts.[564][565]
S0627	SodaMaster	SodaMaster has the ability to download additional payloads from C2 to the targeted system.[203]
S1166	Solar	Solar has the ability to download and execute files.[443]
C0024	SolarWinds Compromise	During the SolarWinds Compromise, APT29 downloaded additional malware, such as TEARDROP and Cobalt Strike, onto a compromised host following initial access.[566]
S0615	SombRAT	SombRAT has the ability to download and execute additional payloads.[161][184][567]
S0516	SoreFang	SoreFang can download additional payloads from C2.[568][569]
S0374	SpeakUp	SpeakUp downloads and executes additional files from a remote server. [570]
S1140	Spica	Spica can upload and download files to and from compromised hosts.[571]
S0646	SpicyOmelette	SpicyOmelette can download malicious files from threat actor controlled AWS URL's.[572]
S0390	SQLRat	SQLRat can make a direct SQL connection to a Microsoft database controlled by the attackers, retrieve an item from the bindata table, then write and execute the file on disk.[573]
S1030	Squirrelwaffle	Squirrelwaffle has downloaded and executed additional encoded payloads.[574][575]
S1112	STEADYPULSE	STEADYPULSE can add lines to a Perl script on a targeted server to import additional Perl modules.[576]
S0380	StoneDrill	StoneDrill has downloaded and dropped temporary files containing scripts; it additionally has a function to upload files from the victims machine.[577]
G1046	Storm-1811	Storm-1811 has used scripted `cURL` commands, BITSAdmin, and other mechanisms to retrieve follow-on batch scripts and tools for execution on victim devices.[578][579][580]
S1183	StrelaStealer	StrelaStealer installers have used obfuscated PowerShell scripts to retrieve follow-on payloads from WebDAV servers.[581]
S1034	StrifeWater	StrifeWater can download updates and auxiliary modules.[582]

ID	Name	Description
S9001	SystemBC	SystemBC has downloaded additional files for execution on the victim's machine.[585][586] The server component of SystemBC has the ability to send additional files to victim machines.[586]
S0663	SysUpdate	SysUpdate has the ability to download files to a compromised host.[445][587]
G1018	TA2541	TA2541 has used malicious scripts and macros with the ability to download additional payloads.[588]
G0092	TA505	TA505 has downloaded additional malware to execute on victim systems.[589][533][590]
G0127	TA551	TA551 has retrieved DLLs and installer binaries for malware execution from C2.[591]
S0011	Taidoor	Taidoor has downloaded additional files onto a compromised host.[592]
S0586	TAINTEDSCRIBE	TAINTEDSCRIBE can download additional modules from its C2 server.[593]
S1193	TAMECAT	TAMECAT has used `wget` and `curl` to download additional content.[414]
S0164	TDTESS	TDTESS has a command to download and execute an additional file.[594]
G1056	TeamPCP	TeamPCP has modified legitimate software binaries to retrieve secondary payloads from C2.[595][596]
S9041	TeamPCP Cloud Stealer	TeamPCP Cloud Stealer has the ability to download additional payloads to targeted systems.[597][598][595][599]
G0139	TeamTNT	TeamTNT has the `curl` and `wget` commands as well as batch scripts to download new tools.[600][601]
S0595	ThiefQuest	ThiefQuest can download and execute payloads in-memory or from disk.[602]
G0027	Threat Group-3390	Threat Group-3390 has downloaded additional malware and tools, including through the use of `certutil`, onto a compromised host .[282][603]
S0665	ThreatNeedle	ThreatNeedle can download additional tools to enable lateral movement.[332]
S0668	TinyTurla	TinyTurla has the ability to act as a second-stage dropper used to infect the system with additional malware.[604]
S0671	Tomiris	Tomiris can download files and execute them on a victim's system.[605]
S1239	TONESHELL	TONESHELL has the ability to download additional files to the victim device.[606]
G0131	Tonto Team	Tonto Team has downloaded malicious DLLs which served as a ShadowPad loader.[607]
S0266	TrickBot	TrickBot downloads several additional files and saves them to the victim's machine.[608][609]
S0094	Trojan.Karagany	Trojan.Karagany can upload, download, and execute files on the victim.[610][611]
G0081	Tropic Trooper	Tropic Trooper has used a delivered trojan to download additional files.[612]
S0436	TSCookie	TSCookie has the ability to upload and download files to and from the infected host.[613]
S9034	Tsundere Botnet	Tsundere Botnet's loader component has downloaded the zip file node-v18.17.0-win-x64.zip from the official Node.js website, as well as pm2, a Node.js process management tool.[614]
S0647	Turian	Turian can download additional files and tools from its C2.[64]

ID	Name	Description
S0130	Unknown Logger	Unknown Logger is capable of downloading remote files.[67]
S0275	UPPERCUT	UPPERCUT can download and upload files to and from the victim's machine.[619][620][621]
S0022	Uroburos	Uroburos can use a `Put` command to write files to an infected machine.[622]
S0386	Ursnif	Ursnif has dropped payload and configuration files to disk. Ursnif has also been used to download and execute additional payloads.[623][624]
S0476	Valak	Valak has downloaded a variety of modules and payloads to the compromised host, including IcedID and NetSupport Manager RAT-based malware.[625][626]
S0636	VaporRage	VaporRage has the ability to download malicious shellcode to compromised systems.[96]
S0207	Vasport	Vasport can download files.[627]
S0442	VBShower	VBShower has the ability to download VBS files to the target computer.[628]
S0257	VERMIN	VERMIN can download and upload files to the victim's machine.[629]
S1217	VIRTUALPITA	VIRTUALPITA has the ability to upload and download files.[630]
G1055	VOID MANTICORE	VOID MANTICORE has deployed additional payloads from dedicated C2 servers.[631][632][633] VOID MANTICORE has also downloaded legitimate tools and software from publicly available services.[631] VOID MANTICORE had utilized VeraCrypt a legitimate disk encrypting utility that was downloaded directly from the website.[631]
G0123	Volatile Cedar	Volatile Cedar can deploy additional tools.[132]
S0180	Volgmer	Volgmer can download remote files and additional payloads to the victim's machine.[634][635][636]
G1017	Volt Typhoon	Volt Typhoon has downloaded an outdated version of comsvcs.dll to a compromised domain controller in a non-standard folder.[637]
S0670	WarzoneRAT	WarzoneRAT can download and execute additional files.[638]
C0037	Water Curupira Pikabot Distribution	Water Curupira Pikabot Distribution used Curl.exe to download the Pikabot payload from an external server, saving the file to the victim machine's temporary directory.[639]
S0579	Waterbear	Waterbear can receive and load executables from remote C2 servers.[640]
S0109	WEBC2	WEBC2 can download and execute a file.[641]
S0515	WellMail	WellMail can receive data and executable scripts from C2.[642]
S0514	WellMess	WellMess can write files to a compromised host.[27][643]
S0689	WhisperGate	WhisperGate can download additional stages of malware from a Discord CDN channel.[644][645][646][647]
G0107	Whitefly	Whitefly has the ability to download additional tools from the C2.[648]
S0206	Wiarp	Wiarp creates a backdoor through which remote attackers can download files.[649]
G0112	Windshift	Windshift has used tools to deploy additional payloads to compromised hosts.[650]

ID	Name	Description
G1035	Winter Vivern	Winter Vivern executed PowerShell scripts to create scheduled tasks to retrieve remotely-hosted payloads.[654]
S1115	WIREFIRE	WIREFIRE has the ability to download files to compromised devices.[655]
G0090	WIRTE	WIRTE has downloaded PowerShell code from the C2 server to be executed.[656]
G0102	Wizard Spider	Wizard Spider can transfer malicious payloads such as ransomware to compromised machines.[657]
S1065	Woody RAT	Woody RAT can download files from its C2 server, including the .NET DLLs, `WoodySharpExecutor` and `WoodyPowerSession`.[658]
S0341	Xbash	Xbash can download additional malicious files from its C2 server.[659]
S0653	xCaon	xCaon has a command to download files to the victim's machine.[97]
S0658	XCSSET	XCSSET downloads browser specific AppleScript modules using a constructed URL with the `curl` command, `https://" & domain & "/agent/scripts/" & moduleName & ".applescript`.[660]
S1248	XORIndex Loader	XORIndex Loader has been used to download a malicious payload to include BeaverTail.[78]
S0388	YAHOYAH	YAHOYAH uses HTTP GET requests to download other files that are executed in memory.[661]
S0251	Zebrocy	Zebrocy obtains additional code to execute on the victim's machine, including the downloading of a secondary payload.[662][125][663][23]
S0230	ZeroT	ZeroT can download additional payloads onto the victim.[664]
S0330	Zeus Panda	Zeus Panda can download additional malware plug-in modules and execute them on the victim's machine.[665]
S1114	ZIPLINE	ZIPLINE can download files to be saved on the compromised system.[655][110]
G0128	ZIRCONIUM	ZIRCONIUM has used tools to download malicious files to compromised hosts.[666]
S0086	ZLib	ZLib has the ability to download files.[384]
S0672	Zox	Zox can download files to a compromised machine.[277]
S0412	ZxShell	ZxShell has a command to transfer files from a remote host.[667]
S1013	ZxxZ	ZxxZ can download and execute additional files.[89]

Mitigations

ID	Mitigation	Description
M1037	Filter Network Traffic	Use network filtering to block outbound traffic from compromised systems to unapproved external destinations. Restricting access to known, trusted IP addresses and protocols can prevent attackers from downloading malicious tools or payloads onto compromised servers after gaining initial access.
M1031	Network Intrusion	Network intrusion detection and prevention systems that use network signatures to identify traffic for specific adversary malware or unusual data transfer over known protocols like FTP can be used to mitigate activity at the

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0060	Detect Ingress Tool Transfers via Behavioral Chain	AN0165	Unusual or uncommon processes initiate network connections to external destinations followed by file creation (tools downloaded).
		AN0166	Shell-based tools (curl, wget, scp) initiate connections to external domains followed by creation of executable files on disk.
		AN0167	Process execution of curl or wget followed by a network connection and a file created in temporary or user-specific directories.
		AN0168	Command line interface or vCLI triggers remote transfer using wget or curl, writing files into datastore paths or local tmp directories.
		AN0169	Network device logs show anomalous inbound file transfers or uncharacteristic flows with high payload volume to network devices with storage or automation hooks.

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